

PROJECT REPORT



Personal Expense Tracker Pro

A Desktop Expense Management Application built with Python & CustomTkinter

 **Income & Expenses**

 **Visual Pie Chart Reports**

 **CSV-Based Storage**

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Domain	Python Programming
Internship Organization	SoftGrowTech
Project Type	Desktop GUI Application

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1 Introduction

The Personal Expense Tracker Pro is a modern desktop application developed using Python and CustomTkinter. The application helps users record and organize their daily income and expenses in an easy and efficient way. It stores all transaction data in a CSV file, calculates financial summaries automatically, and presents spending information through an interactive pie chart.

This project demonstrates practical implementation of Python programming concepts, GUI development, file handling, data processing, and data visualization.

2 Objectives

The main objectives of this project are:

- ✓ Develop a user-friendly desktop application.
- ✓ Record income and expense transactions.
- ✓ Store transaction history permanently.
- ✓ Calculate total income, expenses, and remaining balance.
- ✓ Display expense distribution using graphical visualization.
- ✓ Improve Python GUI programming skills.

3 Technologies Used

- | | |
|--------------------|----------------------|
| ◆ Python 3 | ◆ CustomTkinter |
| ◆ Pandas | ◆ Matplotlib |
| ◆ CSV File Storage | ◆ Tkinter MessageBox |
| ◆ Datetime Module | |

4 Features

The application provides the following features:

- | | |
|--|------------------------------------|
| ◆ Modern CustomTkinter graphical interface | ◆ Add income transactions |
| ◆ Add expense transactions | ◆ Select expense categories |
| ◆ Automatic date generation | ◆ Transaction history table |
| ◆ Delete selected transaction | ◆ Clear all transactions |
| ◆ Automatic balance calculation | ◆ Income and expense summary cards |
| ◆ Expense distribution pie chart | ◆ CSV-based data storage |
| ◆ Input validation | ◆ Success and error notifications |

5 Working Procedure

1

Select Transaction Type

The user selects the transaction type (Income or Expense).

2

Enter Details

The user selects a category and enters a description and amount.

3

Validate Input

The application validates the entered information.

4

Save Transaction

The transaction is saved into the CSV file.

5

Refresh Table

The transaction table is refreshed automatically.

6

Update Summary

Income, expenses, and balance are updated instantly.

7

View Chart

The user can view an expense distribution pie chart.

6 Project Structure

The project follows a clean and organized directory structure:

```
KashafWaheed_PythonProgramming/  
├── expense_tracker.py  
├── transactions.csv  
├── requirements.txt  
├── README.md  
├── LICENSE  
├── screenshots/  
│   ├── home.png  
│   ├── transactions.png  
│   └── chart.png  
└── KashafWaheed_PythonProgramming_Report.pdf
```

7 Modules Used

Module	Purpose
customtkinter	Modern GUI
pandas	Read and write CSV data
matplotlib	Pie chart visualization
tkinter.messagebox	User notifications
datetime	Current date generation
os	File handling

8 Output

The application successfully:

- ✓ Records income and expenses.
- ✓ Stores transaction data permanently.
- ✓ Displays transaction history.
- ✓ Calculates financial summaries.
- ✓ Shows expense analysis using a pie chart.

- ✓ Provides a clean and interactive desktop interface.

9 Learning Outcomes

During the development of this project, the following concepts were learned:

- | | |
|---------------------------------------|----------------------------------|
| ◆ Python GUI development | ◆ Event-driven programming |
| ◆ CSV file handling | ◆ Data manipulation using Pandas |
| ◆ Data visualization using Matplotlib | ◆ Form validation |
| ◆ Desktop application design | ◆ Project organization |
| ◆ GitHub project management | |

10 Conclusion

The Personal Expense Tracker Pro successfully fulfills its objective of providing an easy-to-use desktop application for personal finance management. The project combines graphical user interface design, data storage, financial calculations, and visualization into a practical application.

It demonstrates the knowledge and programming skills gained during the SoftGrowTech Python Programming Internship and serves as a strong portfolio project for future opportunities.